Quiz-2

31st March 2026

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

Optical Remote Sensing

Time:45 minutes *Total Marks: 25*

Section-1 (1 mark= 2)

1. Can two surfaces have the same albedo but appear different in color? Explain how?
2. In the standardised precipitation index (SPI), we transform the rainfall data from gamma distribution to normal distribution. Why?

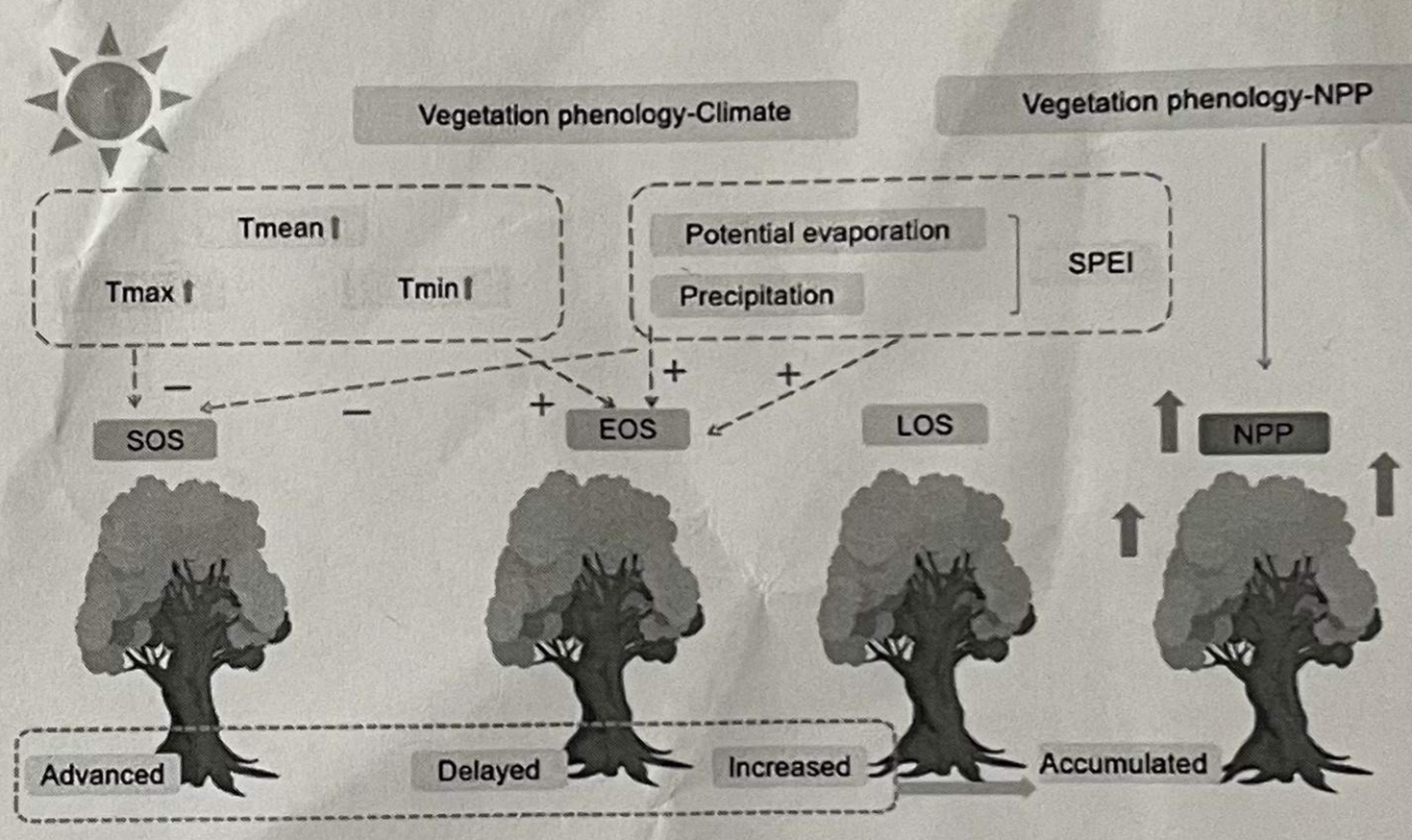
Section-II (2 marks =6)

1. List the inputs required for the development of Biological Richness map of an area
2. What is the role of thermal remote sensing in wildlife monitoring (2 relevant points)
3. In habitat suitability modeling studies (e.g., tiger or red panda), multiple environmental and anthropogenic variables are used. Why is it necessary to include both biophysical variables (e.g., elevation, forest cover) and human-related variables (e.g., distance to settlements)? Explain with reasoning

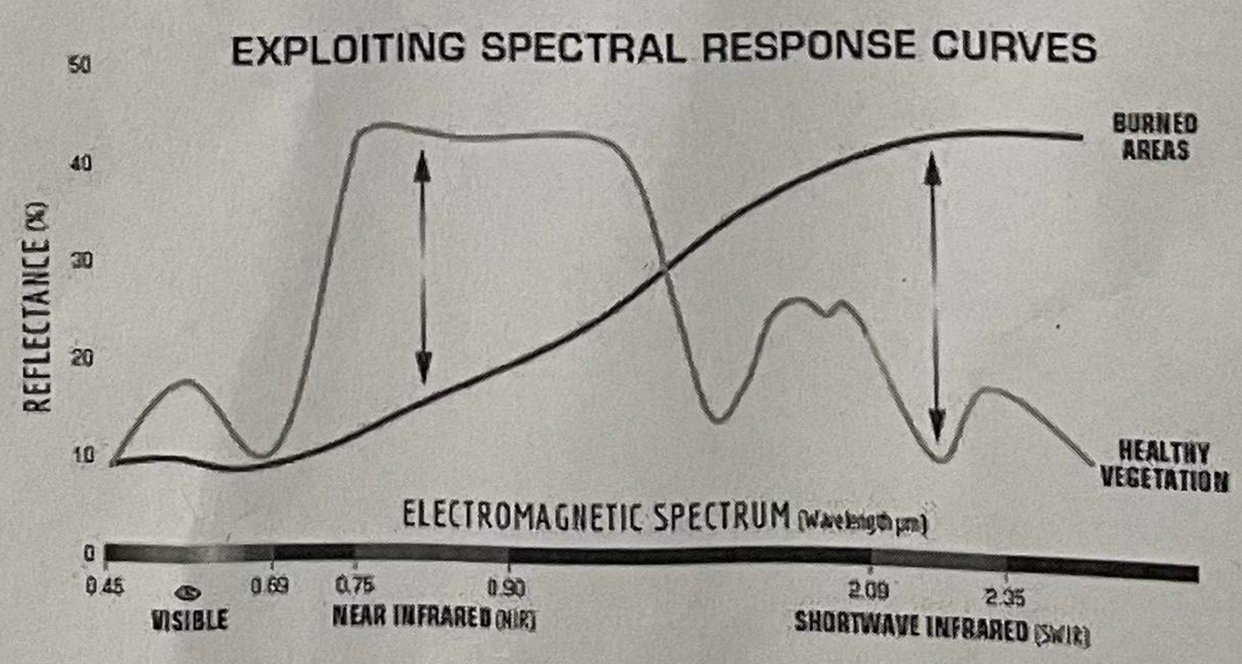
Section-III (3 marks=12)

1. How is EVI estimated? Explain the terms of EVI that give advantage over NDVI in the in the context of crop biomass monitoring?

2. Explain the figure: (only points discussed in the class)



3. I have two rice crop fields, one with sufficient nutrients and the other with nitrogen deficiency. How would the relative NDVI values of the crop fields be? Explain why?
4. Derive three relevant indices from the following figure (Index name and formula)



Section-IV (5 Marks=5)

1. In the context of measuring canopy level LAI (field level), explain the two approaches discussed in class Leaf level
canopy level